Shubham Gogri

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Profile Summary

ML Engineer with Experiences developing ML Systems for Travel and hospitality industry including Data Pipeline, Model development, validation frameworks covering full End-to-end ML pipelines bridging researched POC to production deployment. I Enjoy Collaborating with Data Scientists and Analysts for defining Products and Features architecting Processes making dormant models alive. MSc in AI with Distinction with Projects in Generative Neural Networks ranked **top 10 Best Dissertation Award**, 2024.

Experience

Machine Learning Engineer — **July,2024- Current**

CloudBeds — *London, UK*

- **Tech Stack**: Python, AWS, Airflow, Docker, Streamlit, Plotly-Dash, GitHub Actions, Terraform, PostgreSQL, Snowflake, PyTests, DataDog
- **Built experimental validation framework and Dashboards** (Streamlit + Plotly) interpreting pricing and optimization algorithms, identifying edge cases affecting 30% of properties with production rollout enabled segmented deployment strategy for 20K+ properties.
- Designed and Enambled solutions with **Explainability** features leveraging **SHAP** analysis serving Explainable AI for Revenue Intelligence Platforms that Boosted Confidence in the Analytics Platform.
- **Designed Data Pipelines with Optimized feature engineering algorithms** processing 50+ Millions of Booking Events into Feature Groups usable for Predictive Modelling. Incorporated **Data Quality and Monitoring alerts via DataDog** within these Pipelines Indicating Distribution Shifts with proactive Re-training Mechanism.
- Designed and implemented MLOPs techniques with Git Versioning, Model Tagging, containerized microservices deployed via CI/CD Github workflows on **AWS (Kubernetes, ECS, Lambda, Batch Inferences)** orchestrated via Airflow DAGs.
- Designed and Developed **Model Serving FastAPIs** hosted on Elastic Container Services maintained via **Terraform** leveraging Blue-Green Deployment Strategies and Rollouts.
- **Developed model drift detection system** for time-series forecasting models, implementing novel monitoring approach beyond traditional drift metrics that ensures production model reliability by detecting distribution shifts and performance degradation.

Data Scientist — **1 month Contract**

Qvantia — *Remote, UK*

- Developed Supervised Machine learning pipeline detecting anomalies in **multivariate** sensor data streams from manufacturing systems with an Ensemble technique based on **Random Forest** and **XGBoost** Algorithms .
- Researched Applications for dimensionality reduction techniques such as **PCA** and **LDA** addressing multicollinearity among sensor inputs.
- Articulated findings with models performances to the stakeholders and was awarded with the **Excellence in AI Collaboration** for active engagements with Domain Experts.

Technical Skills

ML Frameworks/ Lib: Pandas, NumPy, PySpark, TensorFlow, PyTorch, scikit-learn, SciPy, Pydantic, Pandera
MLOps-Infra: AWS SageMaker, ECS, MLflow, Airflow, Docker, ECR, GitHub Actions, CI/CD
Cloud Deployment: AWS (Lambda, ECS, S3, Batch, CodeArtifacts, ELB), Terraform (IaC), SageMaker Feature Stores
Languages: Python, SQL, Java
Data Engineering: Redshift, PostgreSQL, Snowflake
Data Viz and Dashboard: Tableau, PowerBI, Dash (Plotly) streamlit

Projects

Deep Learning for Image correction in sparse-view CT | *Tensorflow, Computer Vision, Medical Imaging* **Link**

- Reduced X-ray dosage requirements through Generative Adversarial Networks for reconstruction of sparse-view CT scans and achieved publication recognition with abstract acceptance at **ICMLMI,2023 (International Conference on Machine Learning in Medical Imaging** Link
- Implemented Training Pipelines with Experimental Design Framework including Evaluation Workflows, Tensorboard based Model Performance monitoring and Visualisations.
- Researched Custom Perceptual Loss , Wasserstein Loss functions with Varied Training Methodologies and Pix2Pix architectures achieving 76 Peak Signal to Noise Ratio and 1 Structural Similarity Index Measure (SSIM).

Emotion Classification | *Tensorflow, NLP, Deep Learning, MLOps* **Link**

- Designed **multi-class text classification** pipeline identifying 27 emotion labels in Reddit comments using GoEmotions dataset, achieving competitive performance on fine-grained emotion recognition task
- Trained and evaluated deep learning models including **DistilBERT**, **Bi-LSTM**, and custom **CNN-LSTM hybrid**, experimenting with hyperparameters to analyze trade-offs between model depth and generalization.
- Tracked experiment metrics and configurations using **Weights & Biases (wandb)** for iterative model comparison and performance monitoring.
- Deployed model on web application using Gradio hosted on Hugging Face Spaces.

Education

University of Surrey **Feb. 2023 – Feb 2024**
Masters of Science in Artificial Intelligence *Grade – Distinction*

University of Mumbai **June. 2018 – June 2022**
Bachelors of Engineering in Information Technology *Grade – 8.5/10*

Recognition / Extracurricular

Amongst Top 10 out of 300 for ”Best Dissertation Project”– By the Electronic Engineering Industrial Advisory Board MSc Project Prize by University of Surrey.
”Excellence in AI Collaboration” – SetSquare Surrey’s Entrepreneurship programme (IKEEP and ITeK) for exceptional industry collaboration, communication abilities.
2nd prize ”Generative AI Hackathon”. – Extensive solution tackling Generative AI use in Assessments for Ethical AI Applications.
ITeK– Entrepreneurship and Intrapreneurial Training– Surrey University’s Enterprise cube (SetSquared).